SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I A.Shanbagam /192424260160 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 - AT3

CLOUD COMPUTING

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DATE : 13/10/26.

SHORT ANSWERS TEST

1. Cloud Computing and explain any four characteristics that make it different from traditional on-premise computing
2. Trace the evolution of Cloud Computing from hardware evolution to Internet Software evolution, and show how these changes enabled modern Cloud platform
3. Explain the role of Server Virtualization in Cloud Computing Differentiate between Virtualization as an enabling Virtualization as an enabling technology and Cloud Computing as a Services model
4. Discuss the role of data centers in delivering Cloud Services
5. Differentiate Iaas, Paas, Saas, and Xaas with one Suitable real world example for each model.

1. Cloud computing :-

• The Cloud computing of the four characteristic that maks it different form of traditional on-premise computing.

• Cloud computing is a Computing technology that provides Servers, Storage, network, Software, Internet and database.

Example: Google Drive, where we can store files online

four characteristics:-

- On demand self-services: users can access computing resources whenever in the internet surrounded area.
- It store the database, Google Drive files in the online. Services.
- Whenever in the Internet surrounded area.
- Based network: Clould computing services can be accessed through the internet using mobile laptops and computers
- Resource pooling:- Computing resource are shared among the many users.
- Rapid elasticity: Resources can be increased or decreased according to the user needs.

Diffrence from Traditional computing

In the traditional Computing, Companies maintain their own computers and servers. In Cloud computing the users access resources through the internet without needing to own all the hardware.

can store files online.

rs can access.

n Internet surrounded

Drive files

nded area.

rvice

2) Explain the evolution of Cloud Computing

The evolution of Cloud Computing are there is three stages

1. Hardware Large :-

Large computers were used

2. Virtualization :-

It has multiple virtual machines ran on one server

3. Internet :-

It computers become connected world wide

4. Online Software :-

These computers become connected worldwide and also Applications become available through the Internet

5 Cloud computing :-

It Storage, Servers, and Software become available online.

for example :-

AWS, Microsoft Azure.

3) Server Virtualization:

Server Virtualization is a technology that allows multiple virtual servers to run on a single physical server.

Example:

One physical server can run multiple virtual machines using virtualization software.

Role of Server Virtualization.

1. Reduces hardware costs
2. Improves resource utilization
3. Allows multiple operating system to run on one physical machine
4. Makes server management easier.

Virtualization	Cloud computing
It is a technology	It is a service model
It creates virtual machine	It provides computing services through the internet
It can work without internet based cloud servic	It generally use a network to deliver services
Example :- VMware.	Example :- Amazon web Services (AWS

Virtualization is an important technology that helps cloud computing work efficiently.

11. A data center is a place containing servers, storage, devices, and networking equipments.

Role of Data centers :-

1. Data Storage :- They store large amounts of data

2. Data processing :- They process information and run applications.

3. Networking :- They use security systems and deliver cloud computing services through the networks

4. Security :- They use security systems to protect data

5. Backup and Recovery :- They help restore data when failure occur

6. Availability :- They use backup power and other systems to keep service running.

Example:

AWS data centers and Google data centers.

5) Cloud computing provides different types of services to users.

1. Iaas - Infrastructure as a Service

Iaas provides virtual servers, storage, and networking through the Internet. It can create and manage virtual servers.

Ex: Amazon EC2

2. Paas - platform as a Service.

Paas provides a platform for developing, testing and running applications. It can use to Developers can build application without the managing the underlying

Ex:- Google App Engine.

3. Saas:

It is a Software as a Service (Saas)

It provides Ready to use Software

Ex: Google Docs.

13/10/25

4. Xaas :-

They will Anything / Everything as a Service. It provides various IT services delivered as Services.

Ex:

AWS Lamba.